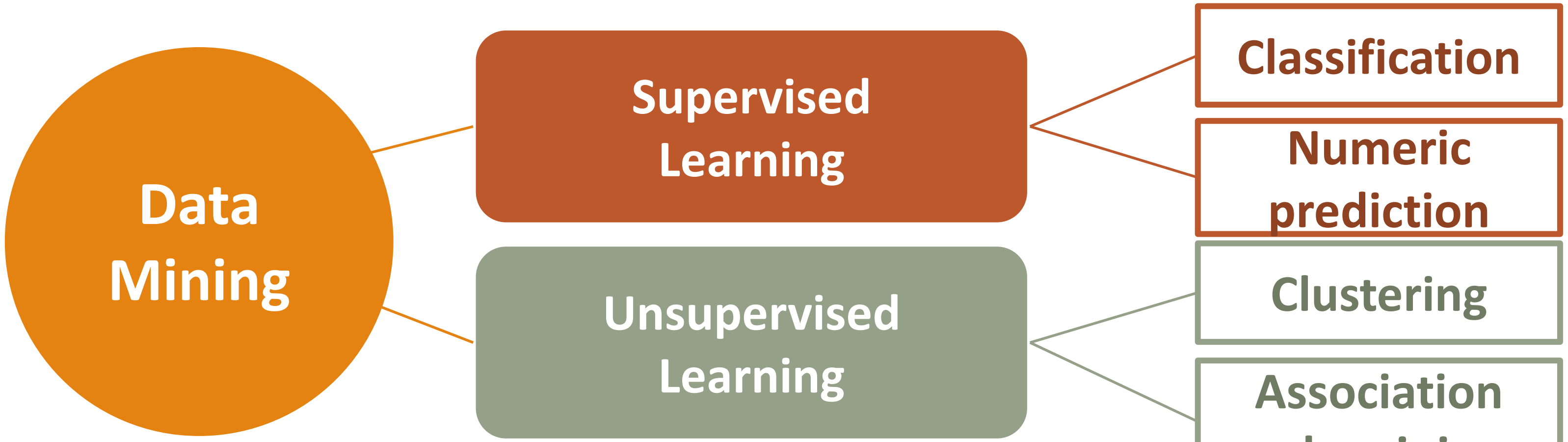
Lecture 11: Improving Model Performance

Data Mining Tasks



Data Mining Tasks

Model	Learning Task
Decision Trees	Classification
Naive Bayes	Classification
k-Nearest Neighbors	Classification
Linear Regression	Numeric Prediction
Regression Trees	Numeric Prediction
Neural Networks	Classification/Numeric Prediction
Support Vector Machines	Classification/Numeric Prediction
K-means clustering	Clustering
Apriori	Association Rule Mining

Supervised Learning

Unsupervised Learning

Outline

- Improve model performance with ensembles
  - Understanding ensembles
  - Voting
  - Bagging
  - Boosting

Improve Model Performance with Ensembles

- Netflix Prize
  - The Netflix Prize was an open competition for the best collaborative filtering algorithm to predict user ratings for films, based on previous ratings.
    - The competition was held by Netflix, an online DVD-rental and video streaming service, and the grand prize is US $1,000,000.
  - Business values:
    - Netflix has around 193 million subscribers globally.
    - Netflix claims its AI assisted recommendation system saves the company $1 billion per year.

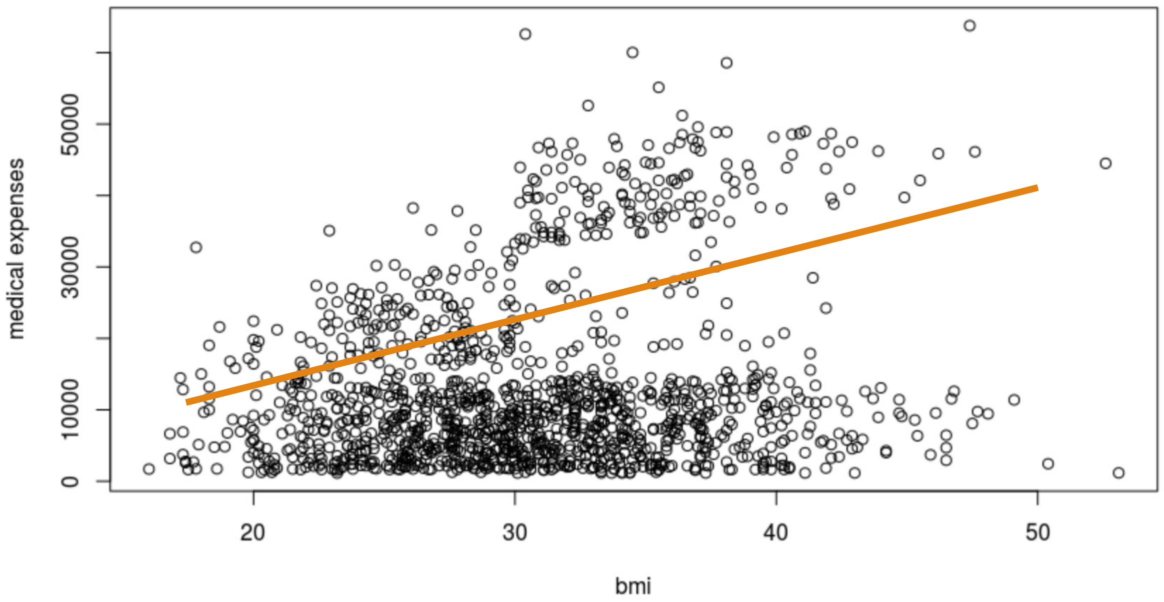
Improve Model Performance with Ensembles

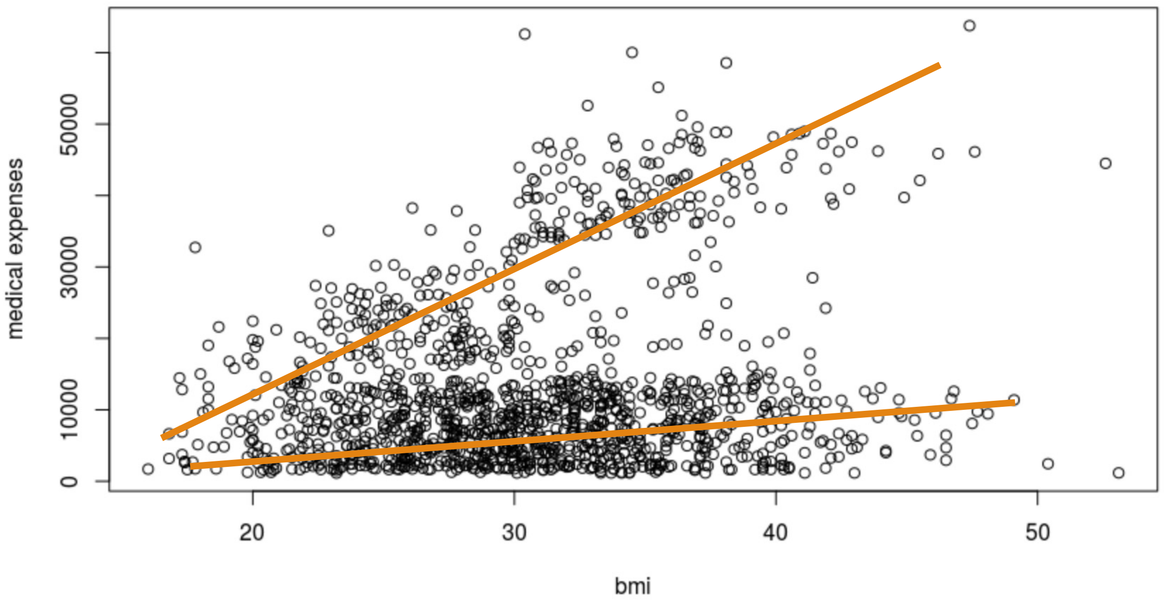
- How can we improve the model performance?

Improve Model Performance with Ensembles

- A model brings a unique bias to a learning task, it may readily learn one subset of examples, but have trouble with another.
  - Highly complex model
  - Ensemble-learning

Improve Model Performance with Ensembles

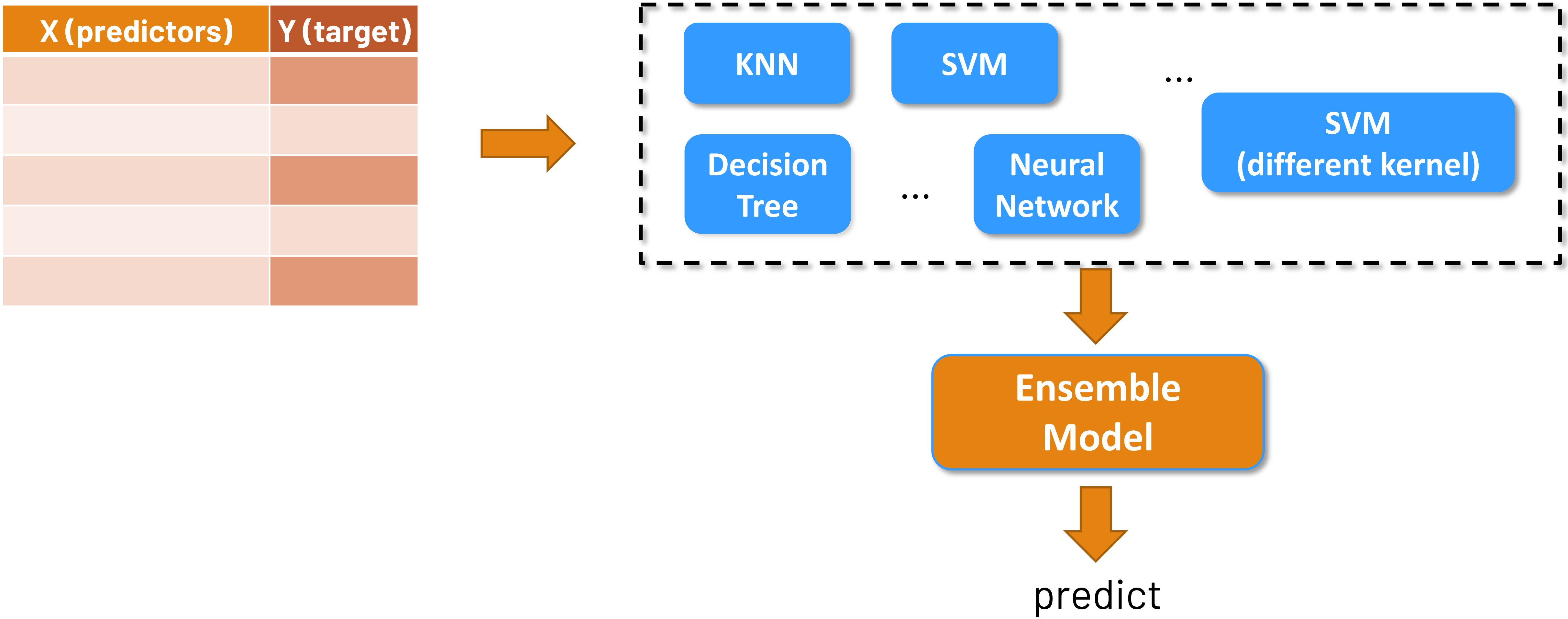




Improve Model Performance with Ensembles

Ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone.

Improve Model Performance with Ensembles



Improve Model Performance with Ensembles

Voting Ensemble Approach

- A voting ensemble (or a "*majority voting ensemble*") is an ensemble machine learning model that combines the predictions from multiple other models.

Improve Model Performance with Ensembles

10,000 patient records

KNN → no

SVM → yes

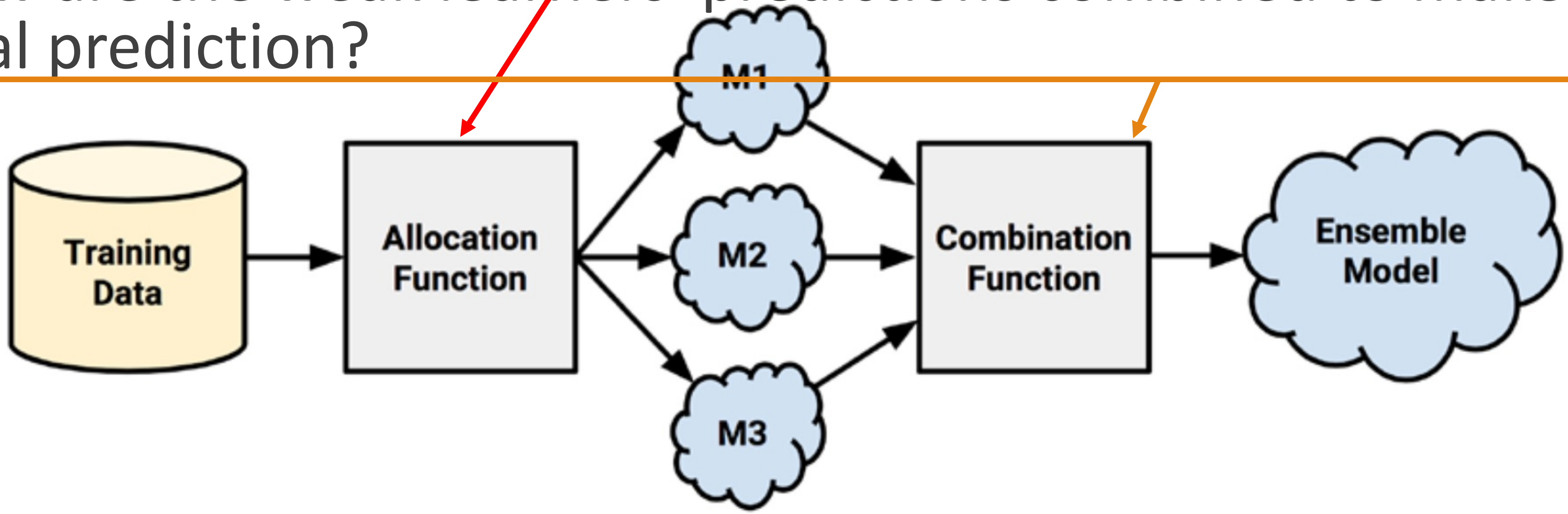
Neural Network → yes

We want to predict whether this patient has lung cancer.

Based on majority rule, we will label the patient outcome as "yes".

Understanding Ensembles

- **Ensemble** method is based on the idea that by combining multiple weaker learners, a stronger learner is created.
  - How are the weak learning models chosen and/or constructed?
  - How are the weak learners' predictions combined to make a single final prediction?



Understanding Ensembles

- Ideal ensemble includes a diverse set of models, the **allocation function** can increase diversity by artificially varying the input data to bias the resulting learners.
- The **allocation function** dictates
  - How much of the training data each model receives.
  - Do they each receive the full training dataset or merely a sample?
  - Do they each receive every feature or a subset?

Understanding Ensembles

- After the models are constructed, they can be used to generate a set of predictions, which must be managed in some way.
- The **combination function** governs how disagreements among the predictions are reconciled.
  - Majority vote
  - Weighting each model's votes
  - Utilize another model to learn a combination function from various combinations of predictions.

Understanding Ensembles

- One of the benefits of using ensembles is that they may allow you to spend less time in pursuit of a single best model.
- Better generalizability to future problems
  - As the opinions of several learners are incorporated into a single final prediction, no single bias is able to dominate.
- Capture subtle patterns that a single global model might miss

Understanding Ensembles

- Ensembles methods: technique of combining and managing the predictions of multiple models.
  - Voting
  - Bagging
  - Boosting

Ensemble Method: Bagging

Ensemble Method: Bagging

- Bagging generates a number of training datasets by bootstrap sampling the original training data.
  - Bagging perform quite well as long as it is used with relatively **unstable** learners
    - Unstable models are essential in order to ensure the ensemble's diversity in spite of only minor variations between the bootstrap training datasets

Ensemble Method: Bagging

- **Bootstrap sampling** refer to the statistical methods of using random samples of data to estimate the properties of a larger set.
  - The creation of several randomly selected training and test datasets, which are then used to estimate performance statistics

Ensemble Method: Bagging

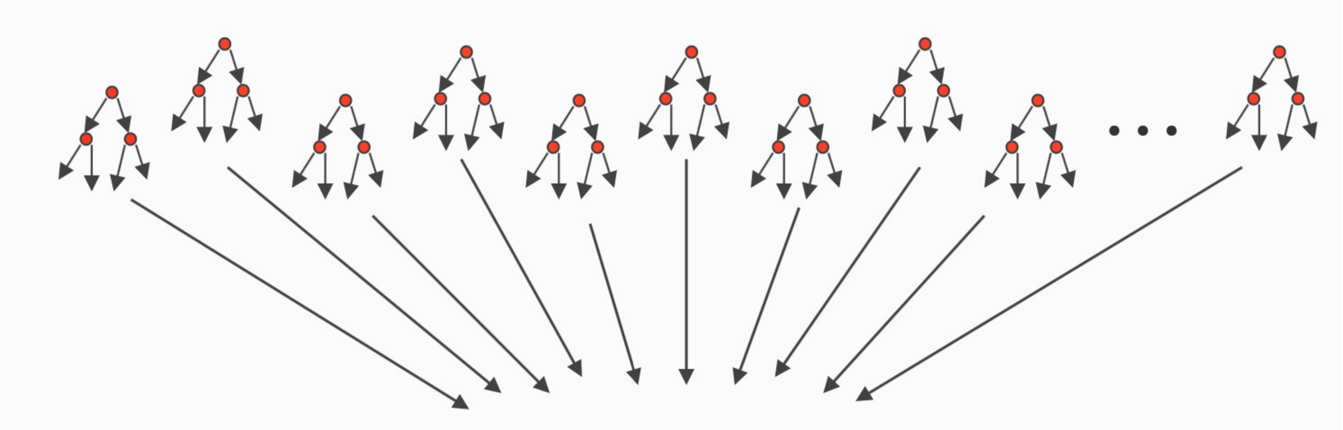
Age	DistToWork	DualInc	Education	Gender	Income	Language	NbrInHouseH	NbrInHouseh	OwnRent	YrsInArea	Rider
7	14	N	3	F	3	English	1	0	Rent	5	Yes
7	10	N	5	M	8	English	2	0	Own	5	No
3	9	N	3	M	1	English	1	0	Rent	5	Yes
1	13	N	2	M	1	English	5	3	Parent	5	Yes
3	14	N	5	F	2	English	3	1	Parent	5	Yes
1	12	N	2	M	1	English	5	0	Parent	5	Yes
2	10	N	4	M	3	English	4	2	Rent	2	Yes
3	13	N	5	M	6	English	1	0	Own	5	No
7	9	Y	3	M	4	English	2	0	Own	4	No
2	10	N	4	F	8	English	2	0	Parent	5	Yes
2	12	N	4	F	2	English	3	0	Parent	5	Yes
2	13	N	3	M	1	English	4	1	Parent	5	Yes
3	12	N	4	F	4	English	4	1	Rent	2	Yes
1	11	N	2	F	1	Other	9	7	Parent	4	Yes
2	13	N	3	M	3	English	1	0	Rent	5	Yes
2	11	N	4	M	2	English	6	0	Parent	5	Yes
4	11	N	3	F	3	English	1	0	Rent	4	No
7	11	N	5	M	8	English	2	0	Own	5	No
6	13	Y	1	F	2	Spanish	4	1	Rent	3	Yes
1	10	N	2	M	8	English	3	1	Parent	5	No
2	13	Y	3	M	6	English	2	0	Own	5	No
6	9	Y	6	M	8	English	9	2	Own	4	No
3	11	N	6	F	4	English	2	0	Parent	4	Yes
3	13	Y	4	F	8	Other	2	0	Own	5	Yes
1	12	N	2	F	1	English	8	6	Parent	5	Yes
1	13	N	2	F	1	English	4	1	Parent	4	Yes
4	9	N	6	F	7	English	2	0	Own	3	Yes

Training data

Testing data

Ensemble Method: Bagging

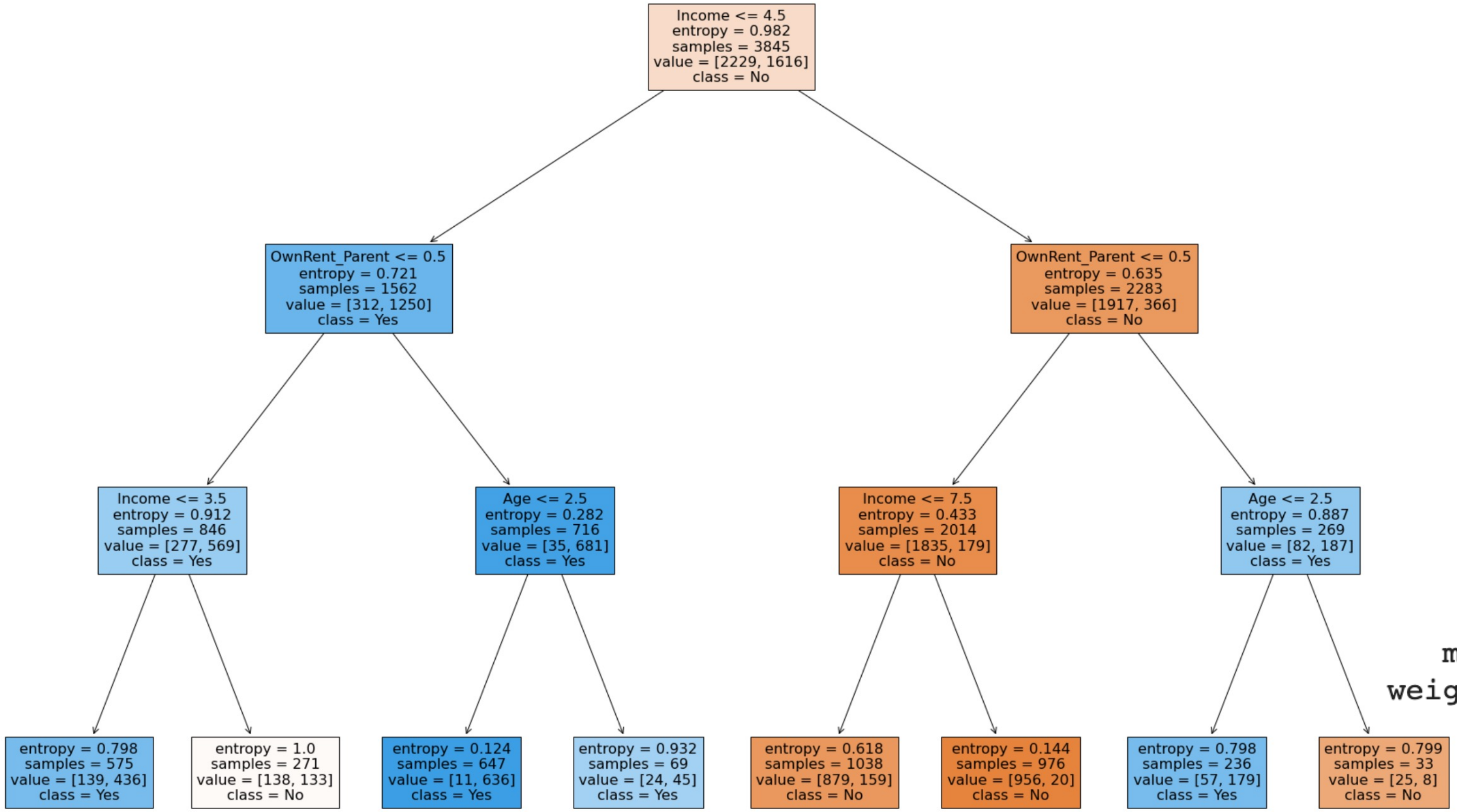
- Bagged Decision Trees
  - Draw T bootstrap samples of data
  - Train trees on each sample
  - Average prediction of trees on out-of-bag samples



Majority voting approach

Ensemble Method: Bagging

```
model = DecisionTreeClassifier(criterion = "entropy", random_state = 1, max_depth = 3)
```



	precision	recall	f1-score	support
No	0.83	0.92	0.87	910
Yes	0.89	0.77	0.83	738
accuracy			0.85	1648
macro avg	0.86	0.85	0.85	1648
weighted avg	0.86	0.85	0.85	1648

Ensemble Method: Bagging

- Bagged Decision Trees
  - model_bagging = BaggingClassifier(base_estimator=DecisionTreeClassifier(criterion = "entropy", random_state = 1, max_depth = 3), n_estimators=20, random_state=1)

```
              precision    recall  f1-score   support

          No       0.89      0.86      0.88       910
         Yes       0.84      0.87      0.85       738

    accuracy                           0.86      1648
   macro avg       0.86      0.86      0.86      1648
weighted avg       0.86      0.86      0.86      1648
```

Ensemble Method: Boosting

Ensemble Method: Boosting

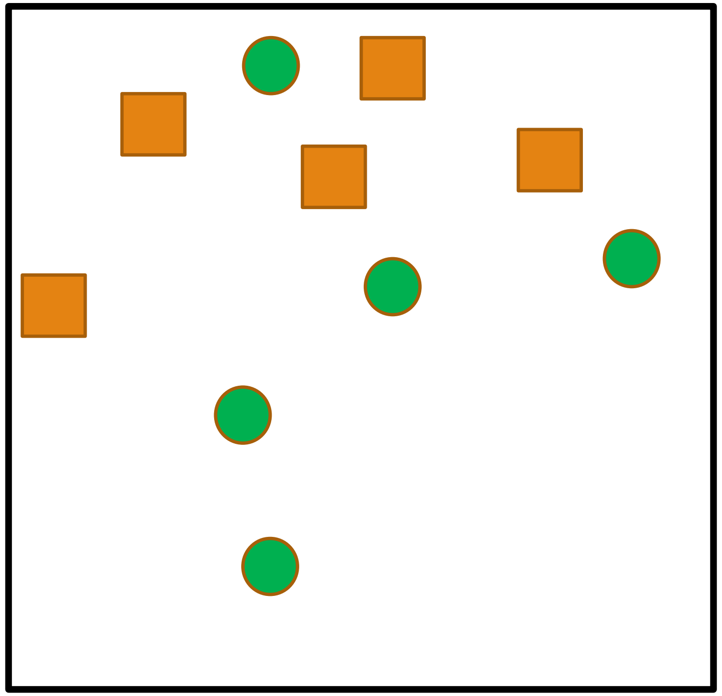
- Another common ensemble-based method is called **boosting** because it boosts the performance of weak learners to attain the performance of stronger learners.

Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**
  - The algorithm is based on the idea of generating weak learners that iteratively learn a larger portion of the difficult-to-classify examples by paying more attention (that is, giving more weight) to frequently misclassified examples.

Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**

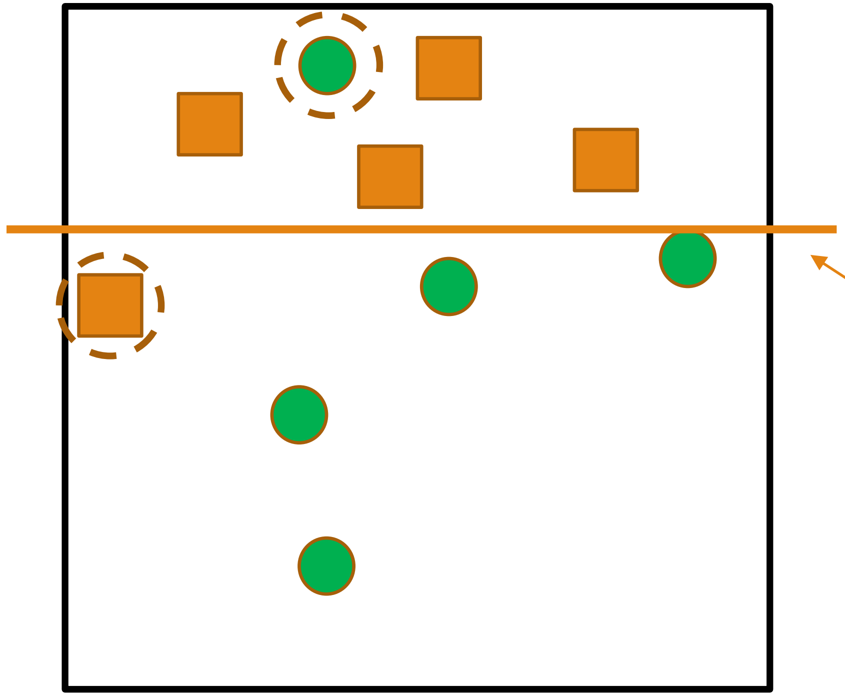


Each example/instance has the same importance

Weak Leaner: an axis parallel line

Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**



Find the best classifier on the dataset

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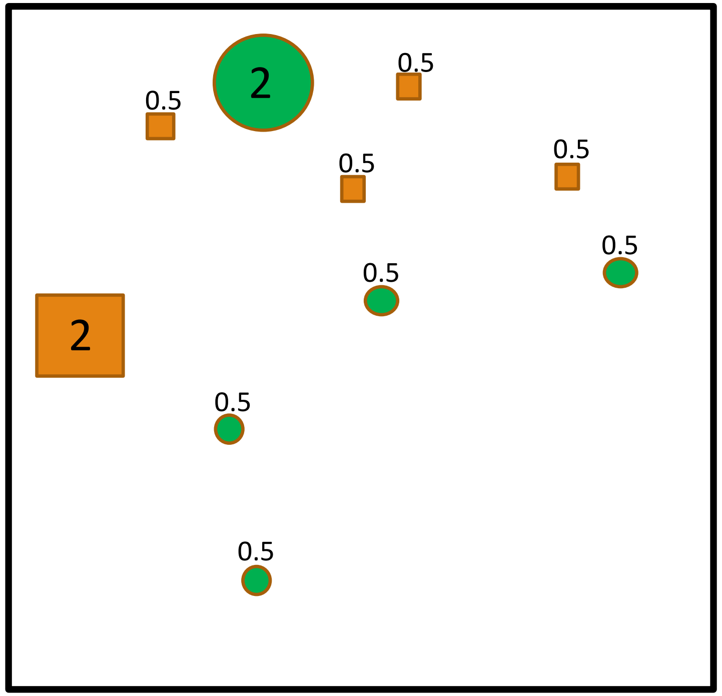
Ensemble Method: Boosting

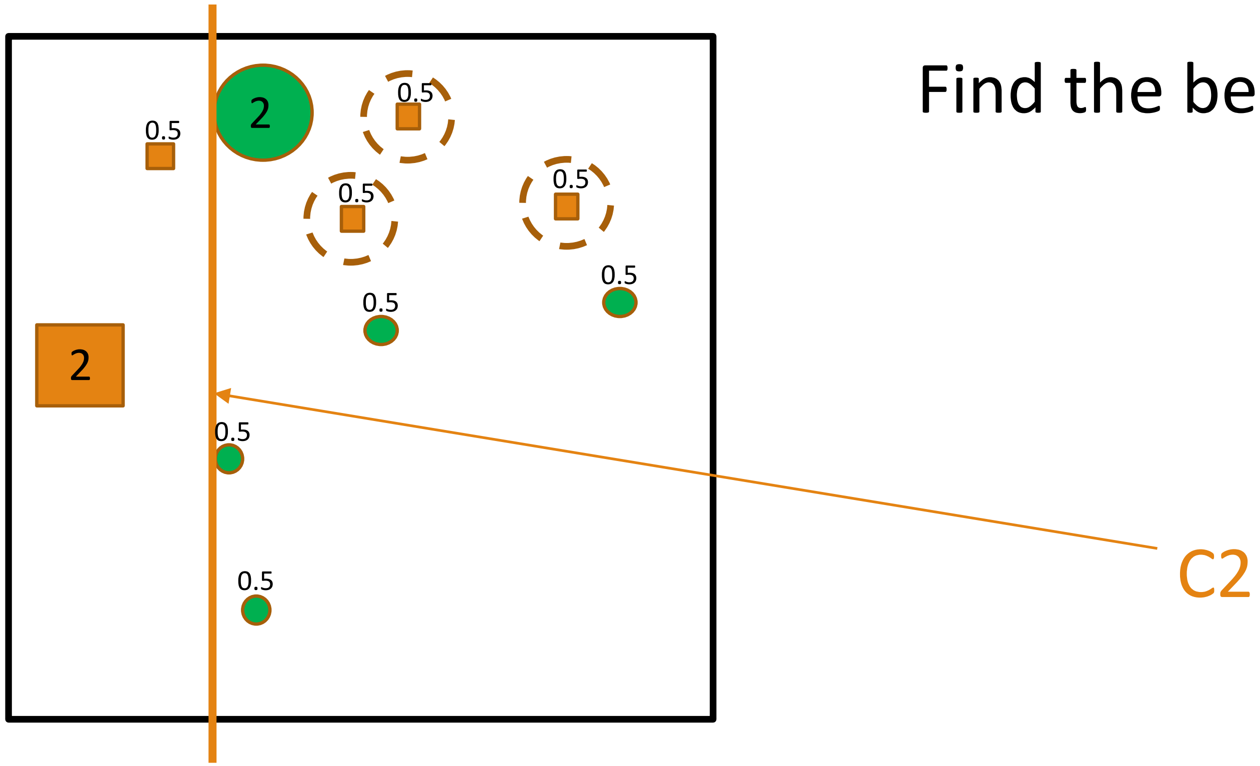
- **AdaBoost** or **adaptive boosting**



Increase the importance of the examples with mistakes and down-weight the examples that got correctly

Ensemble Method: Boosting

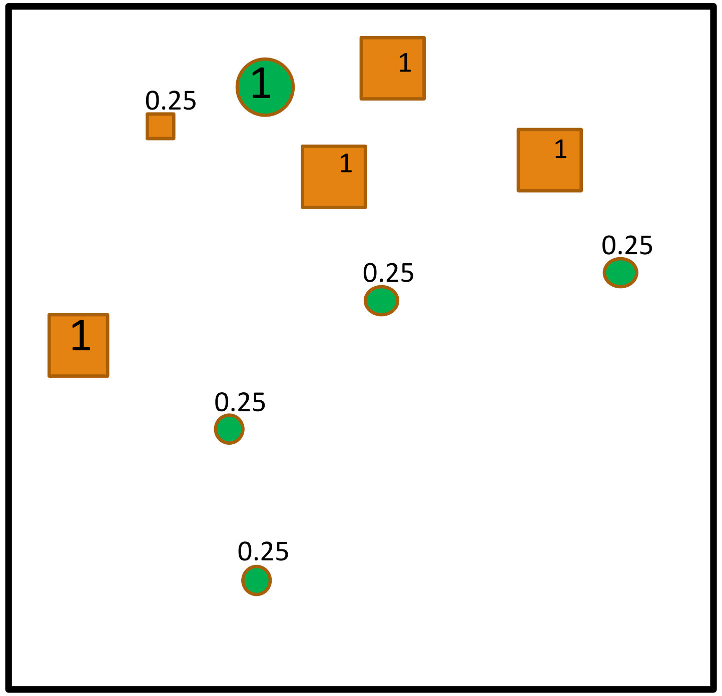
- **AdaBoost** or **adaptive boosting**



Find the best classifier on the dataset

Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**



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Ensemble Method: Boosting

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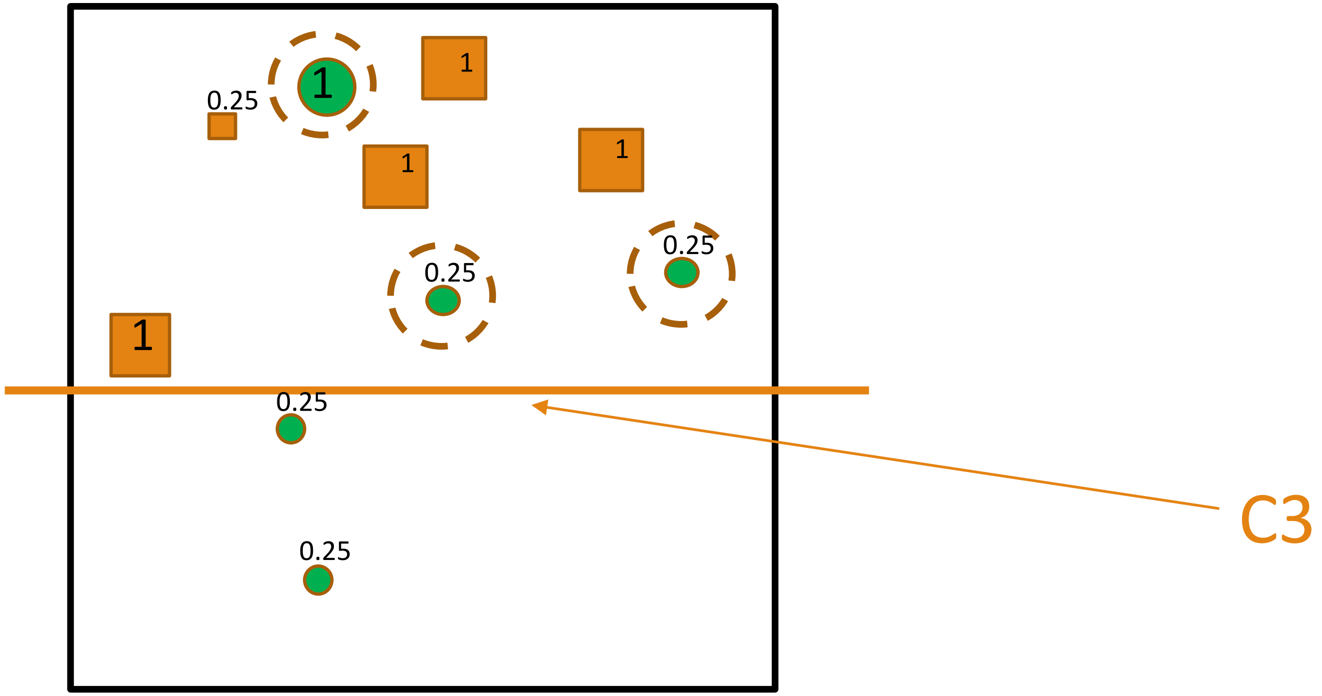
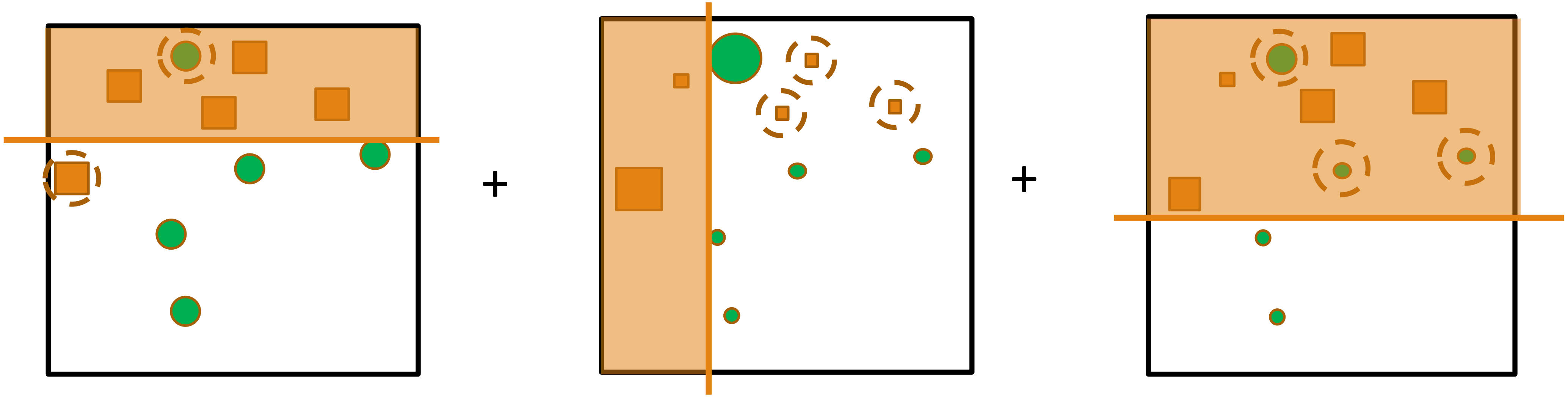


Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**



Ensemble Method: Boosting

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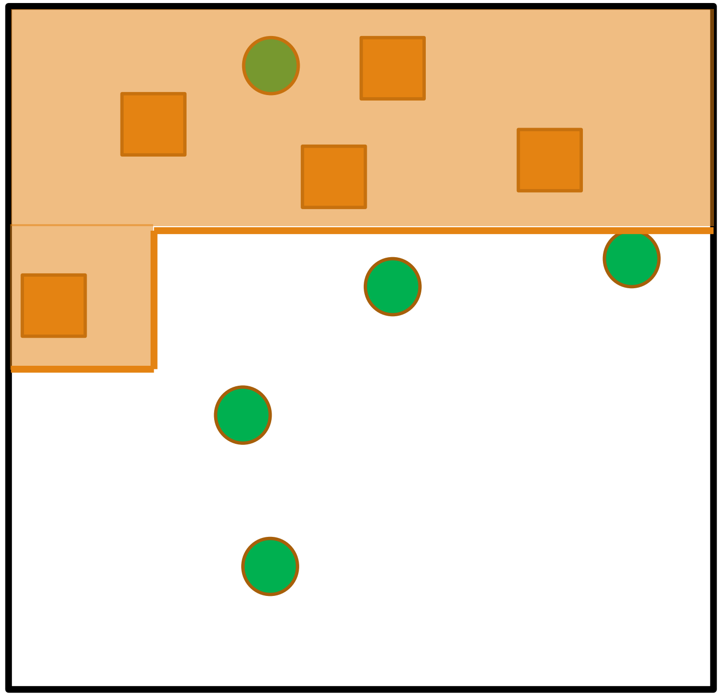

Ensemble Method: Boosting

- Initialization
  - Weigh all training samples equally
- Iteration Steps
  - Train model on (weighted) training set
  - Compute error of model on training set
  - Increase weights on training cases get wrong
- Return final model
  - Carefully weighted prediction of each model

Ensemble Method: Boosting

- **AdaBoost** or **adaptive boosting**
  - model_Boost = AdaBoostClassifier(base_estimator=DecisionTreeClassifier(criterion = "entropy", random_state = 1, max_depth = 3), n_estimators=10, random_state=1)

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